

1. Product Overview:

The WIN-SN-TNH-CO2-M-(OD) is an industrial-grade outdoor environmental monitoring sensor designed for accurate measurement of Temperature, Relative Humidity, and Carbon Dioxide (CO₂) concentration. The device integrates multiple sensing parameters into a single compact unit and communicates through an RS485 interface using the industry-standard Modbus RTU protocol. The sensor is available with CO₂ measurement ranges of 0 to 20,000 / 30,000 / 40,000 ppm, making it suitable for a wide range of indoor and outdoor air quality monitoring applications. Real-time measurements of temperature, humidity, and CO₂ concentration enable effective environmental monitoring, ventilation control, industrial process monitoring, and smart building management.

The WIN-SN-TNH-CO2-M-(OD) supports user-configurable communication settings, including Modbus Slave ID, Baud Rate, Parity, and Stop Bit configuration, allowing seamless integration with PLCs, SCADA systems, HMIs, IoT gateways, data loggers, and Building Management Systems (BMS). Designed for reliable outdoor operation, the sensor features a robust enclosure and stable long-term performance. It is ideal for air quality monitoring, smart buildings, greenhouses, industrial facilities, warehouses, educational institutions, commercial spaces, and environmental monitoring applications.

2. Key Features

- Measures Temperature, Relative Humidity, and Carbon Dioxide (CO₂).
- Industrial-grade sensor designed for outdoor environmental monitoring.
- RS485 communication interface with Modbus RTU protocol support.
- Real-time monitoring and transmission of environmental data.
- User-configurable communication parameters through software settings.
- Configurable Modbus Slave ID (1 to 247).
- Supports baud rates up to 115200 bps.
- Configurable Parity and Stop Bit settings.
- Factory default settings: 9600 bps, Slave ID 1, Parity 3, Stop Bit 1.
- Built-in factory reset functionality for restoring default communication settings.
- High accuracy and long-term measurement stability.
- Fast response time for environmental monitoring applications.
- Supports multi-drop RS485 network architecture.
- Easy integration with PLCs, SCADA systems, HMIs, IoT gateways, and data loggers.

- Robust outdoor enclosure for reliable field deployment.
- Low maintenance and long operational life.
- Suitable for air quality monitoring, smart buildings, greenhouses, industrial automation, warehouses, and environmental monitoring systems.

- Certifications:



Sensor Specification:-

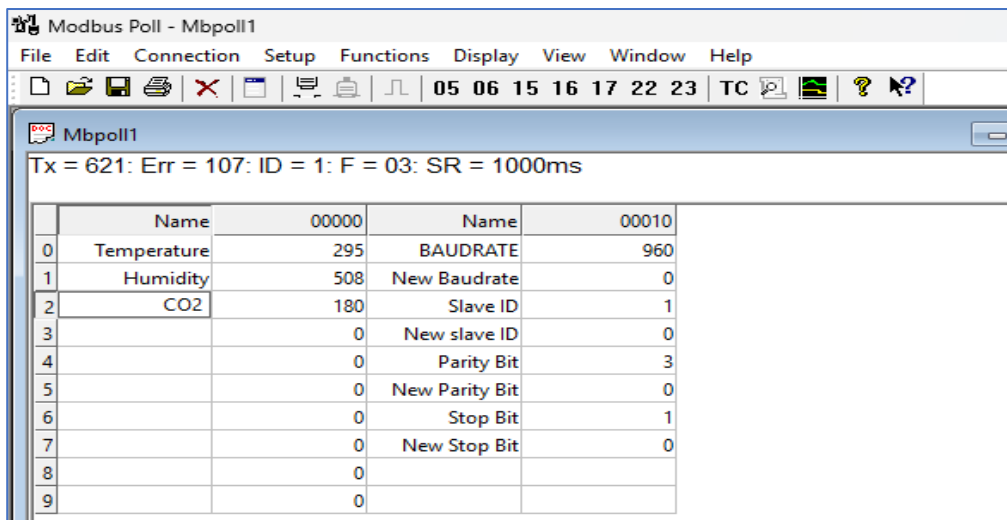
Measuring Parameter	Measuring Range	Accuracy
Temperature	-40 °C to +120 °C	±0.3 °C
Humidity	0 ~ 100 % RH	±2 % RH
CO2	0 – 20000 PPM	-

3. Technical Specifications

- **Operating Voltage:** 12-24v DC
- **Interface:** Modbus RS485
- **Function Code:** 0x03
- **Product dimension (D X H) :** 140 X 200 mm

4. Communication and Configuration Settings:-

The WIN-SN-TNH-CO2-M-(OD) communicates using the RS485 interface with the Modbus RTU protocol. The device is factory-configured with a Baud Rate of 9600 bps, Slave ID 1, Parity setting 3, and 1 Stop Bit, allowing quick deployment in standard Modbus networks.



Modbus Poll - Mbpoll1

File Edit Connection Setup Functions Display View Window Help

05 06 15 16 17 22 23 TC

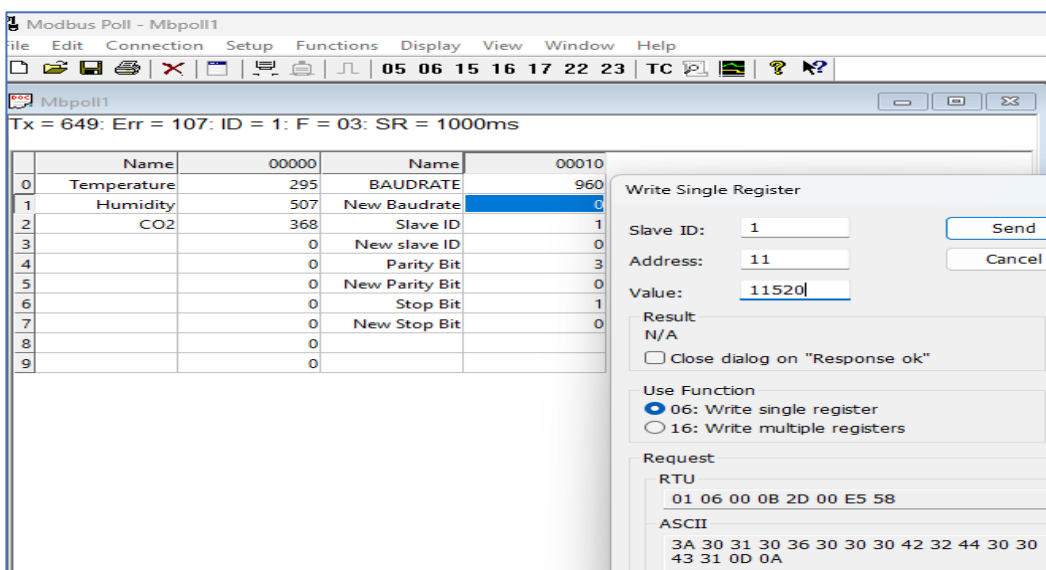
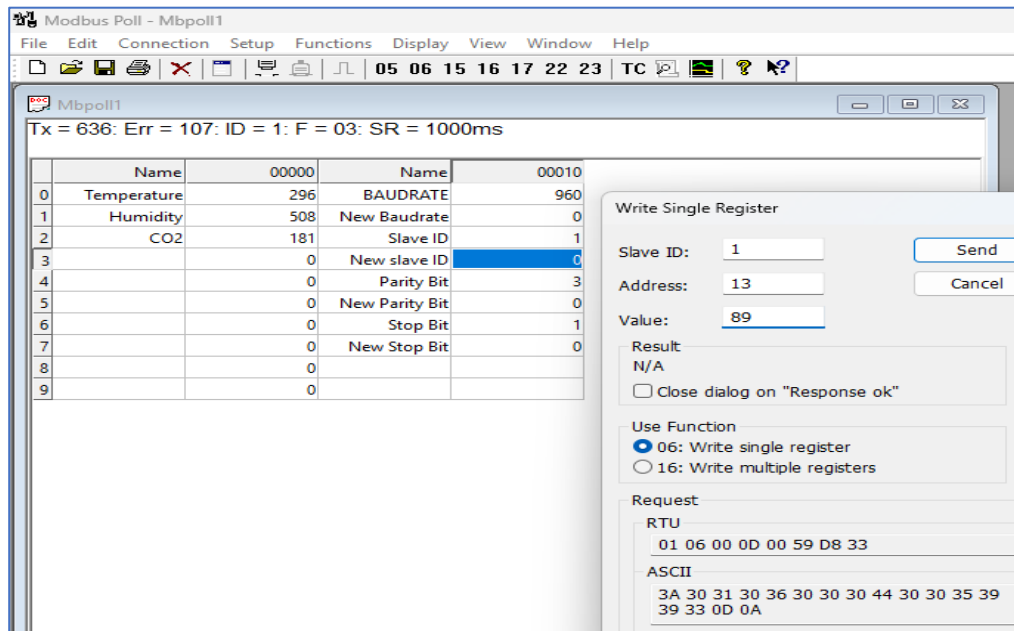
Mbpoll1

Tx = 621: Err = 107: ID = 1: F = 03: SR = 1000ms

	Name	00000	Name	00010
0	Temperature	295	BAUDRATE	960
1	Humidity	508	New Baudrate	0
2	CO2	180	Slave ID	1
3		0	New slave ID	0
4		0	Parity Bit	3
5		0	New Parity Bit	0
6		0	Stop Bit	1
7		0	New Stop Bit	0
8		0		
9		0		

To support a wide range of industrial applications, communication parameters can be modified through software settings. Users can configure the Baud Rate up to 115200 bps, assign a Modbus Slave ID from 1 to 247, and adjust the Parity and Stop Bit settings according to system requirements. These configurable parameters provide flexibility for integration with PLCs, SCADA systems, IoT gateways, HMIs, and other Modbus-compatible devices.

Example screenshots illustrating the software configuration procedure are provided below for reference.



Factory Reset:-

The device includes a factory reset feature that restores all communication parameters to their default settings. To perform a factory reset, place the configuration jumper in the Factory Reset position and press and hold the Reset button for approximately 5 seconds. The device will automatically restart and restore the factory default communication settings:

- Baud Rate: 9600 bps
- Slave ID: 1
- Parity: 3
- Stop Bit: 1

This feature allows users to quickly recover communication with the device if the configured settings are unknown or incorrectly configured.

5. Modbus Register Values :-

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	Temperature	40000	0x03	RS485	16-bit int
2	Humidity	40001	0x03	RS485	16-bit int
3	CO2	40002	0x03	RS485	16-bit int



Note – Comes with base plate for installation on a Pole etc.



Contact Information

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